Bike Share / Trip Data EDA and Insights

Dataset: Divvy Trip Data | **Source:** Kaggle | **Rows/Columns:** **5,667,717 × 13**

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1. Project Objective

- Perform exploratory data analysis (EDA) on Divvy bike-share trips to uncover usage patterns, operational trends, and user behavior.
- Generate actionable insights for bike demand, station usage, and rider type preferences.
- Address specific business questions related to temporal, spatial, and categorical ride attributes.

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2. Data Cleaning and Feature Engineering

Objective: Ensure dataset integrity, standardization, and derivation of analytical features for robust insight generation.

2.1 Data Standardization & Validation
- Harmonized variable types for all columns to their appropriate formats.
- Applied **text normalization**: removed leading/trailing whitespace, enforced consistent capitalization and formatting.
- Identified and addressed **anomalous numeric and temporal records**, including:
- Negative ride durations
- Unrealistic trip distances or durations exceeding operational thresholds (>24 hours)
- Conducted logical consistency checks to ensure all rows conformed to valid operational parameters.

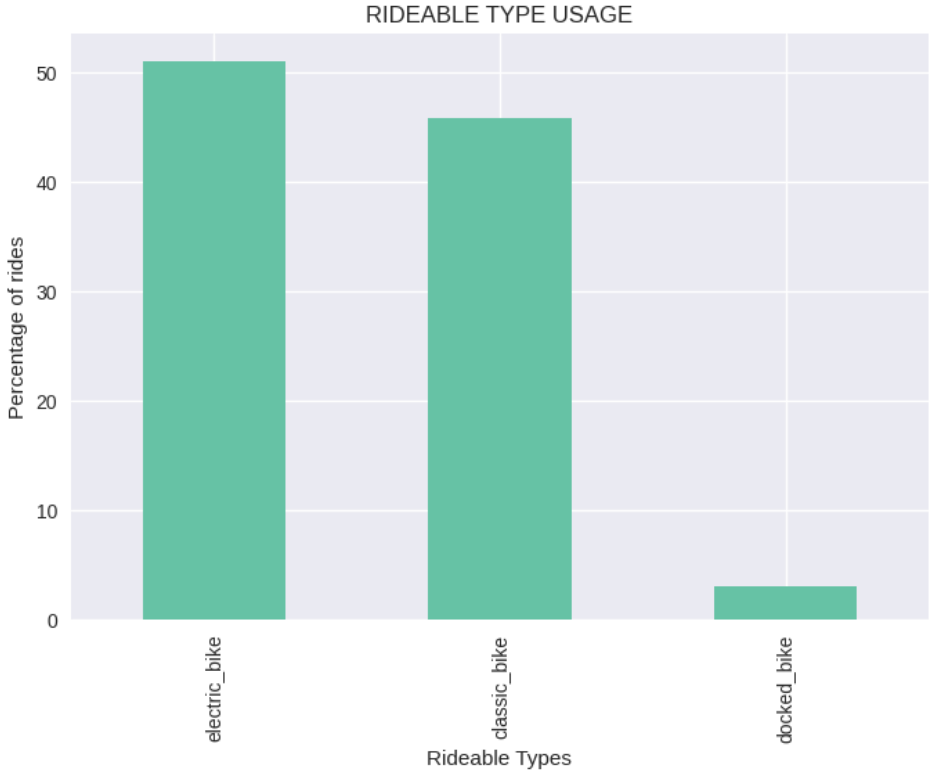
2.2 Feature Engineering
- Derived temporal dimensions:
- **Hour of ride**
- **Day of week**
- **Month**
- **Weekend indicator** (Boolean: True/False)
- Calculated operational metrics:
- **Ride length** (duration in minutes)
- **Trip distance**
- Validated all derived features, ensuring exclusion of extreme outliers or inconsistent entries.

***Outcome**:* Dataset fully cleaned, standardized, and enriched with derived features, providing a robust foundation for EDA and correlation analyses.

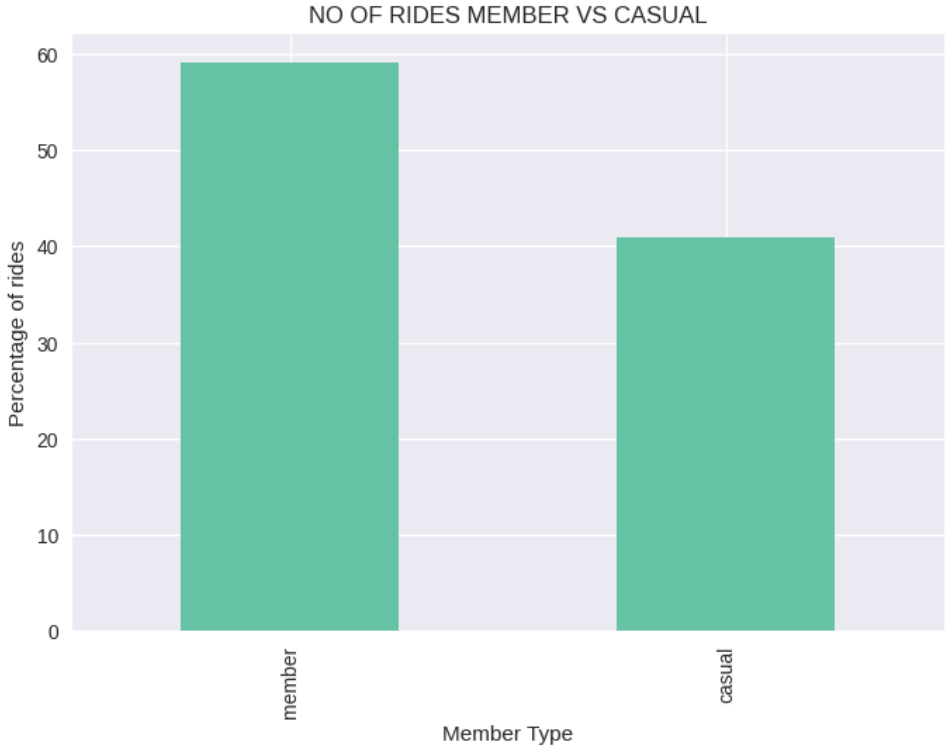
3. Univariate Analysis

Business Questions & Findings:

- **Rideable Types Used:**
  - Electric Bike: 51%
  - Classic Bike: 45.9%
  - Docked Bike: 3.1

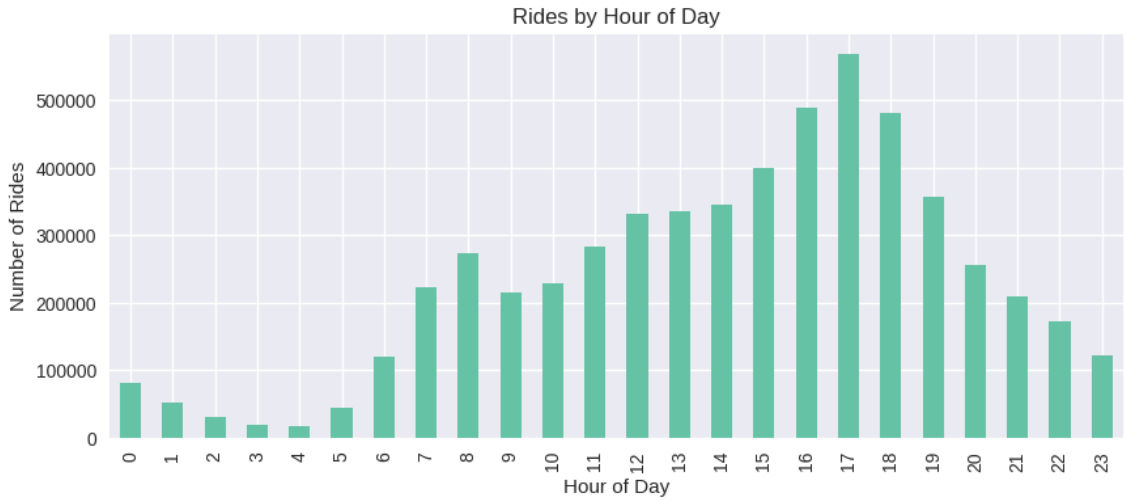


- **User Type Distribution:**
  - Members: 59.1%
  - Casual Riders: 40.9%



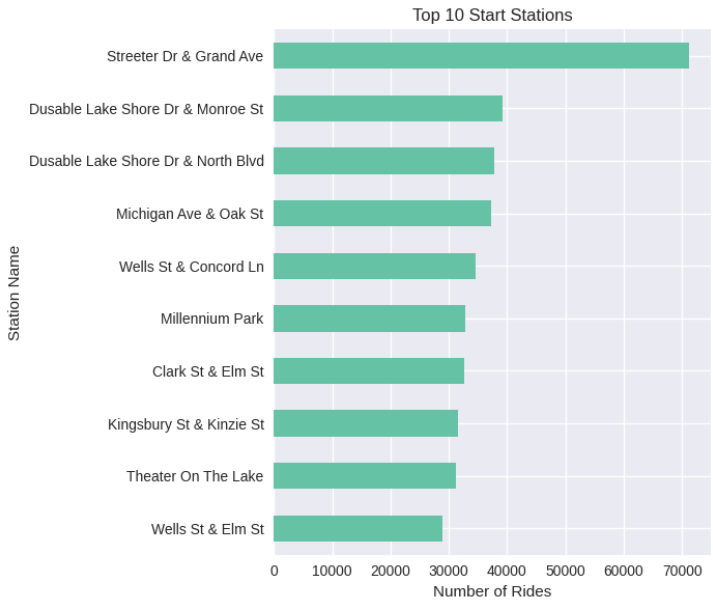
- **Rides by Day of Week:**
  - Highest: Saturday (915,134 rides)
  - Lowest: Monday (750,215 rides)

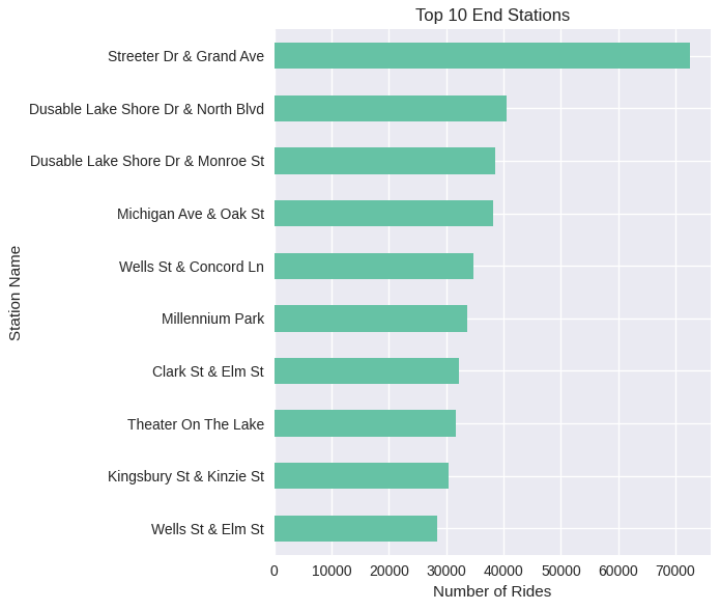
- **Rides by Hour of Day:**
  - Peak usage: 17:00 (569,053 rides)
  - Lowest usage: 4:00 (16,501 rides)



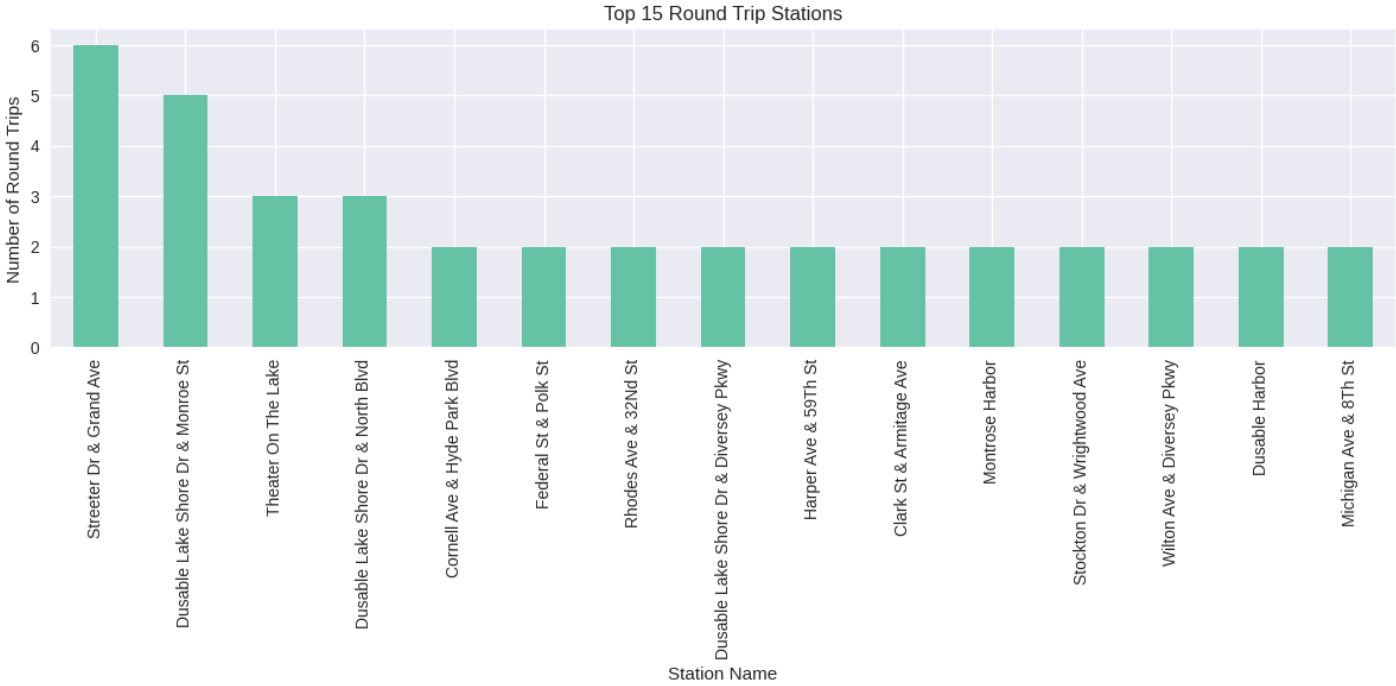
- **Top Start Station:** Streeter Dr & Grand Ave leads (71,255 rides)

- **Top End Station:** Streeter Dr & Grand Ave leads (72,528 rides)

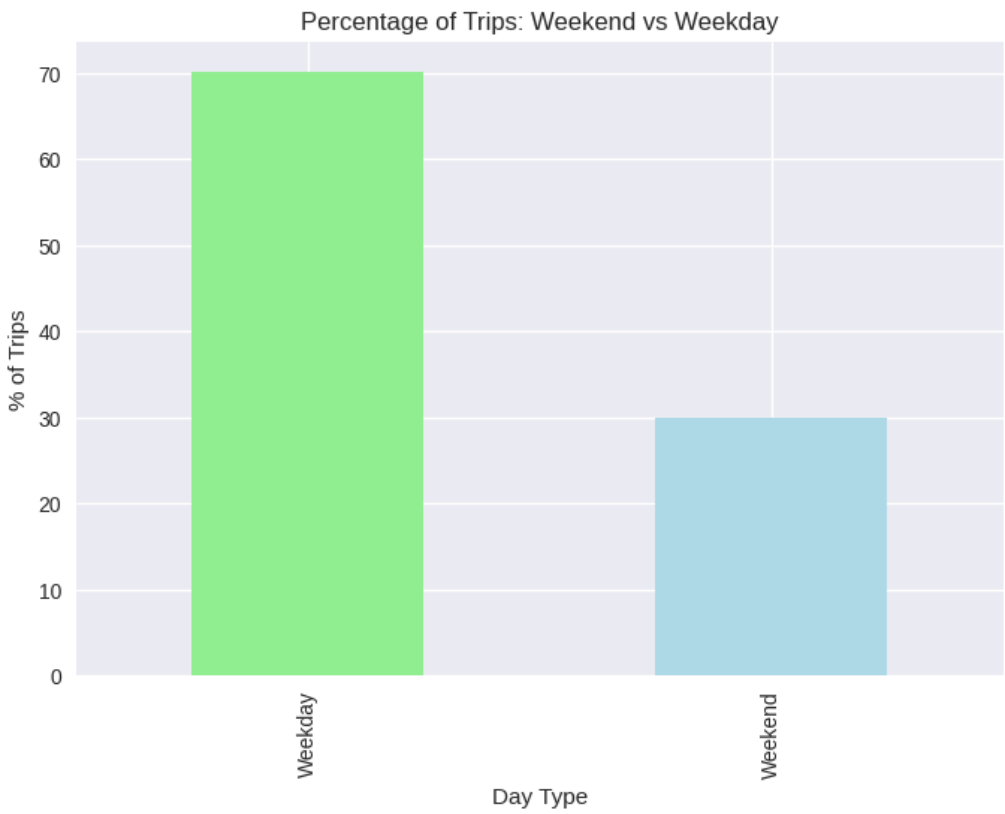




- **Top Round-Trip Station:** Streeter Dr & Grand Ave (6 round trips)



- **Weekend vs Weekday Trips:** Weekdays dominate (70.1%), weekends 29.9%

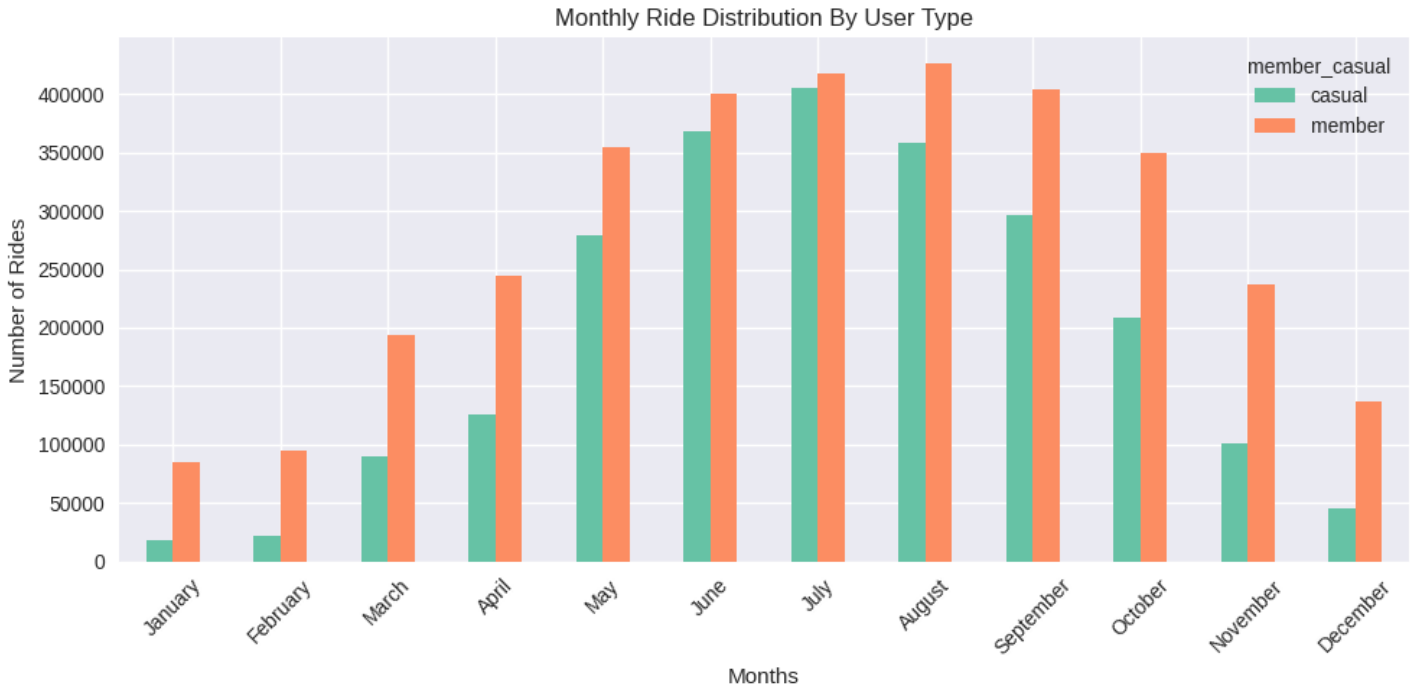


- **Rides by Month:**
  - Peak usage: July (822,445 rides)
  - Lowest: January (103,658 rides)

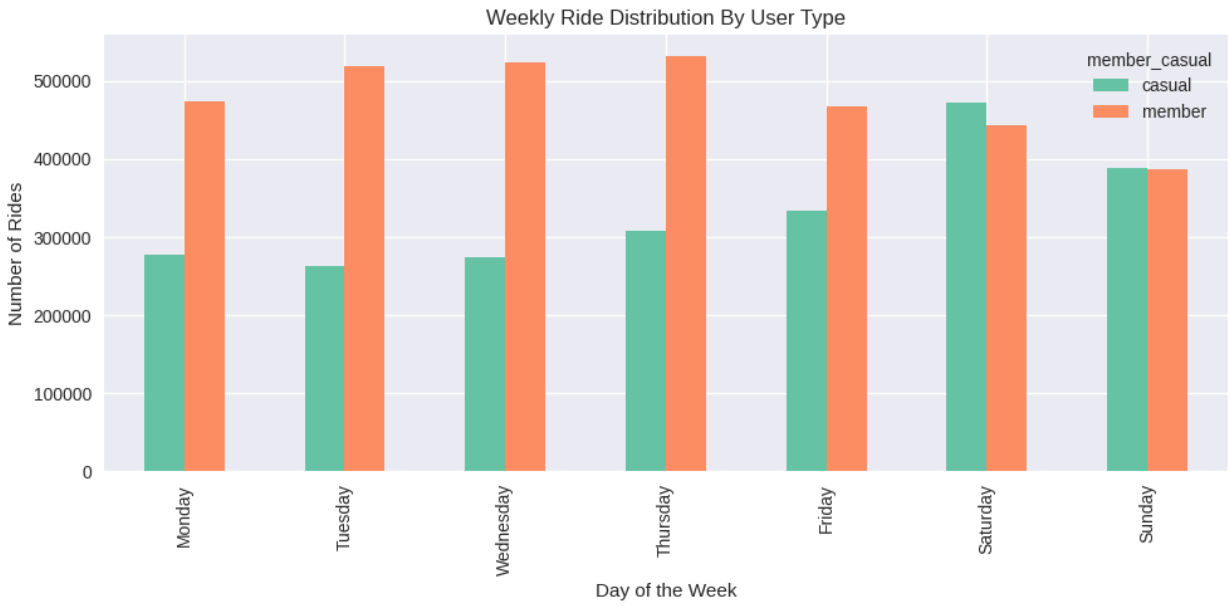
4. Bivariate Analysis

Key Business Questions & Insights:

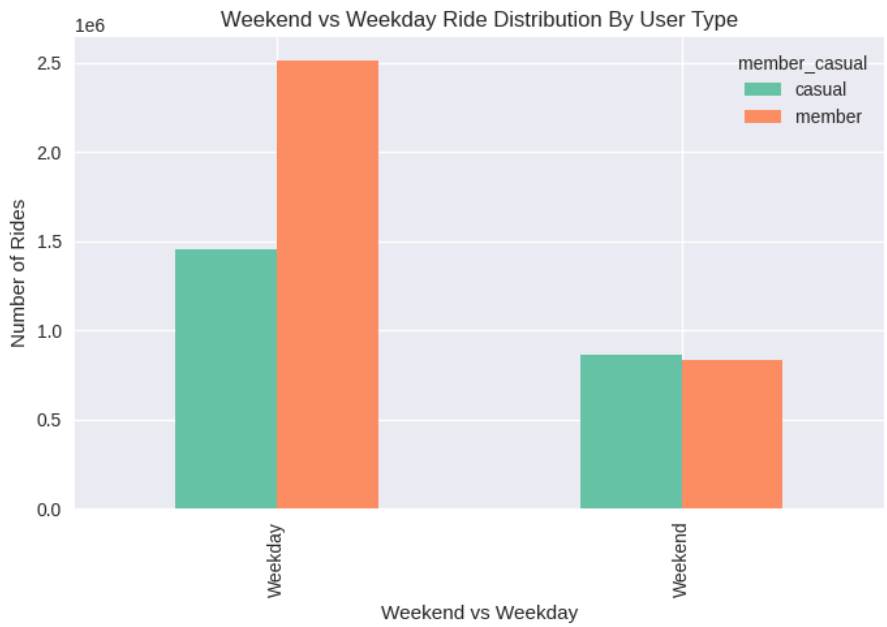
1. **Rider Type vs Month**
   - Casual riders peak in summer months (May–August), members show a steadier pattern.
   - Insight: Casual users are highly seasonal, indicating potential marketing or promotional opportunities.
   - Actionable takeaway: Summer campaigns can boost casual rider engagement, while winter promotions help maintain activity.



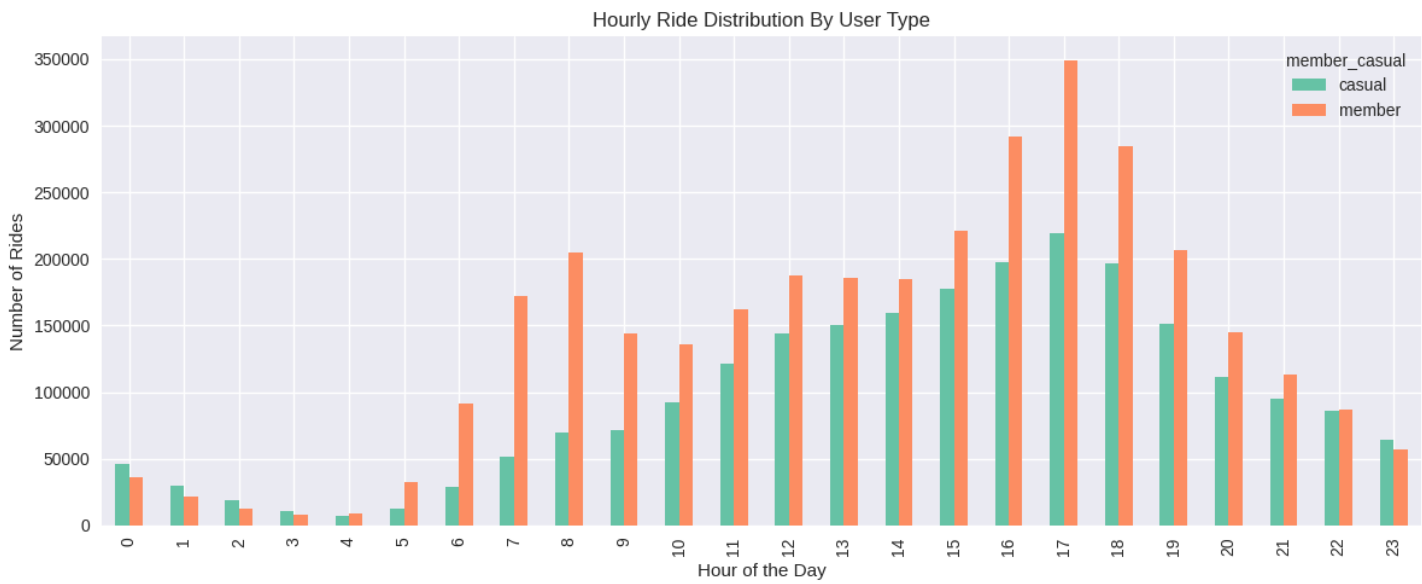
2. **Rider Type vs Weekday**
   - Members dominate weekdays; casual riders dominate weekends.
   - Insight: Members likely use bikes for commuting; casual riders prefer leisure trips.



3. **Rider Type: Weekend vs Weekday**
   - Weekday trips: 2,514,466 (members) vs 1,456,495 (casual)
   - Weekend trips: 830,221 (members) vs 859,940 (casual)
   - Insight: Behavioral segmentation is clear; operational planning can consider this temporal demand split.

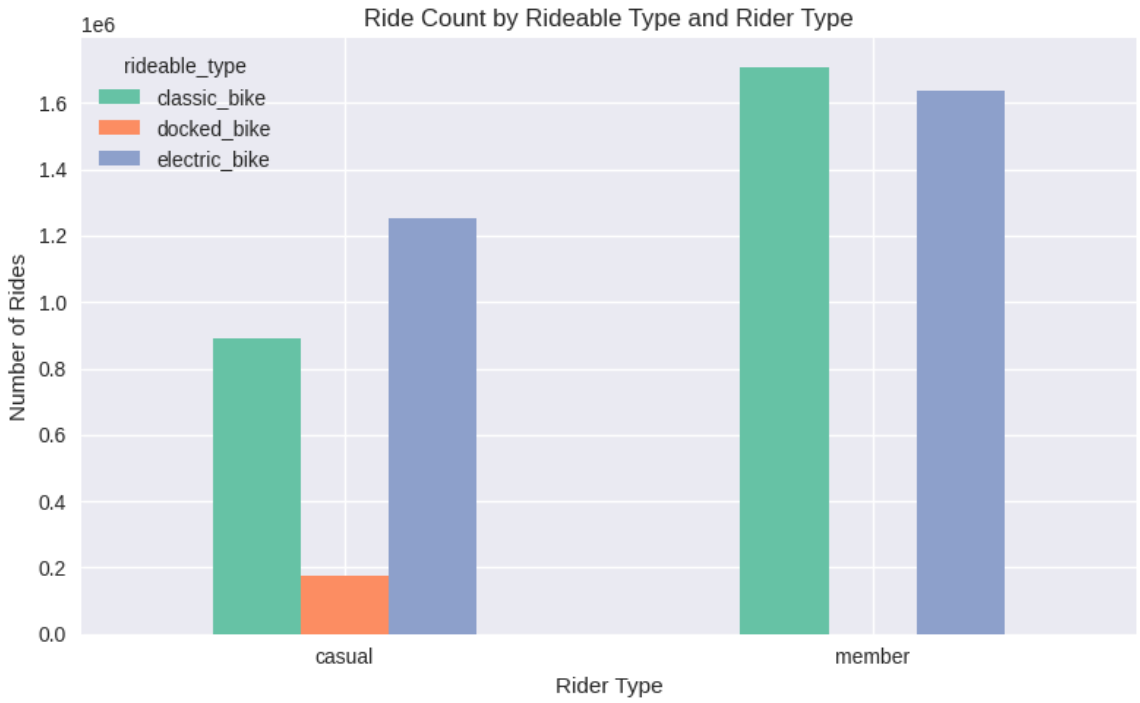


4. **Rider Type vs Hour of Day**
   - Members: peak during commuting hours (6–9 AM, 16–18 PM)
   - Casual: peak in afternoons/evenings (11–20 PM)
   - Insight: Hourly bike redistribution can optimize availability.



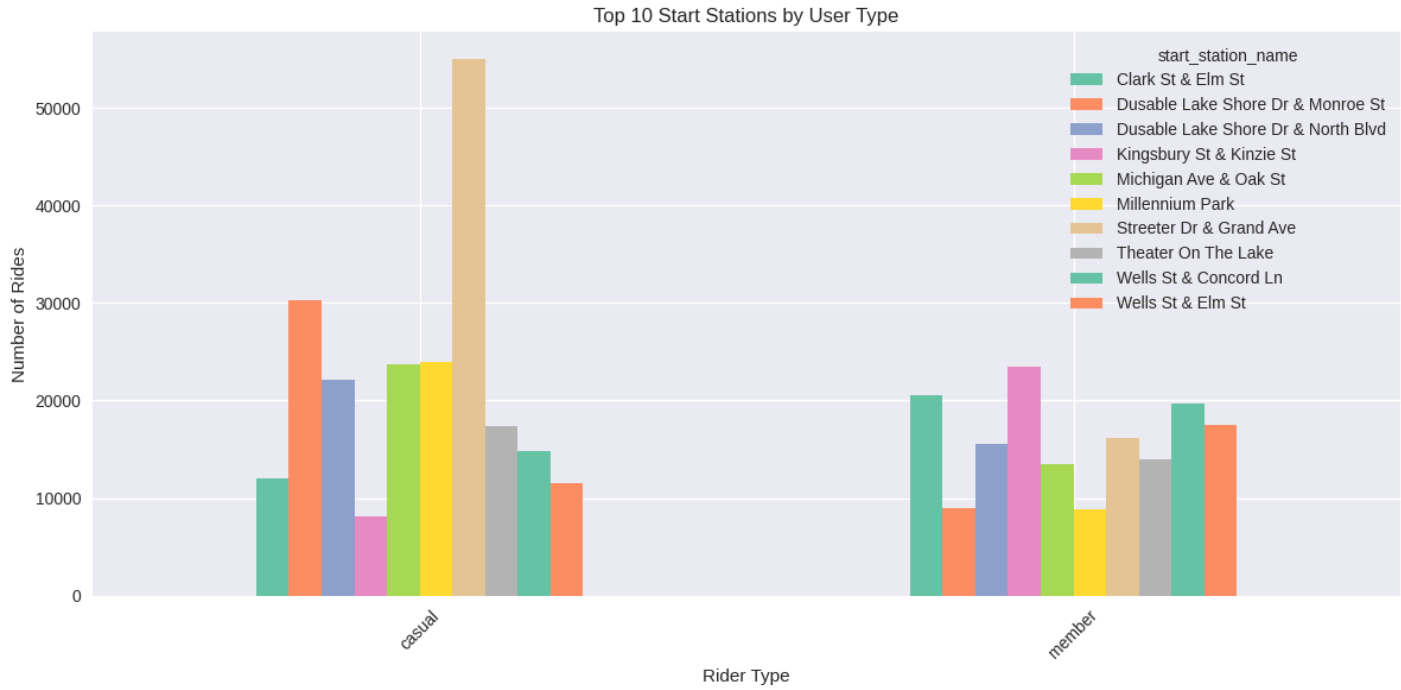
5. **Rider Type vs Ride Length**
   - Average ride: Members 12.4 min, Casual 21.8 min
   - Insight: Casual rides are longer, likely for recreational purposes.

6. **Rider Type vs Rideable Type**
   - Casual riders prefer electric bikes (1,252,895 rides)
   - Members use classic bikes more (1,708,952 rides)
   - Insight: Bike type demand varies by user segment; fleet management can reflect these preferences.

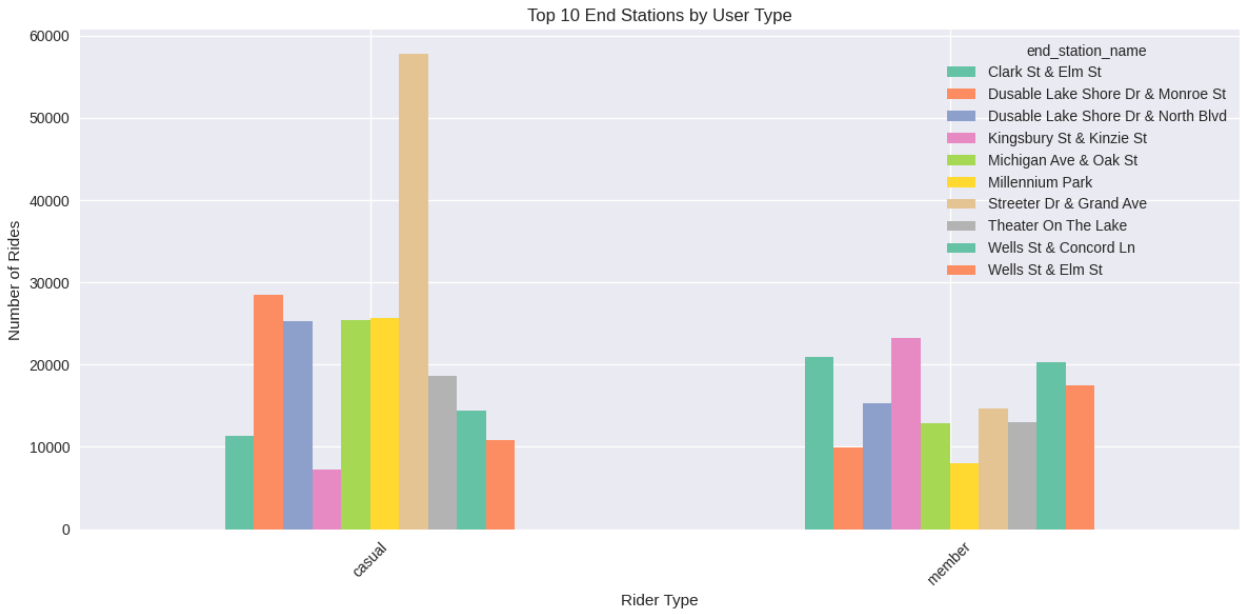


7. **Rider Type vs Top Stations & Insights:**

- **Top 10 Start Stations:**
  - **Casual riders:** Streeter Dr & Grand Ave leads with 55,050 rides; other popular stations include Dusable Lake Shore Dr & Monroe St (30,256) and Dusable Lake Shore Dr & North Blvd (22,156).
  - **Members:** Clark St & Elm St leads with 20,576 rides; Kingsbury St & Kinzie St (23,523) and Michigan Ave & Oak St (13,444) are also frequently used.



- **Top 10 End Stations:**
  - **Casual riders:** Streeter Dr & Grand Ave is again highest (57,801 rides), followed by Millennium Park (25,672) and Michigan Ave & Oak St (25,371).
  - **Members:** Clark St & Elm St is most popular (20,898 rides), with Kingsbury St & Kinzie St (23,213) and Wells St & Concord Ln (20,272) also high.



- **Pattern & Insight:**
  - Casual riders tend to concentrate on **central/high-traffic recreational stations**, while members show a **more dispersed commuting pattern**.
  - **Actionable takeaway:** Allocate bikes based on **station-level user type demand**, prioritizing high-traffic casual stations on weekends and member-heavy stations during weekdays.

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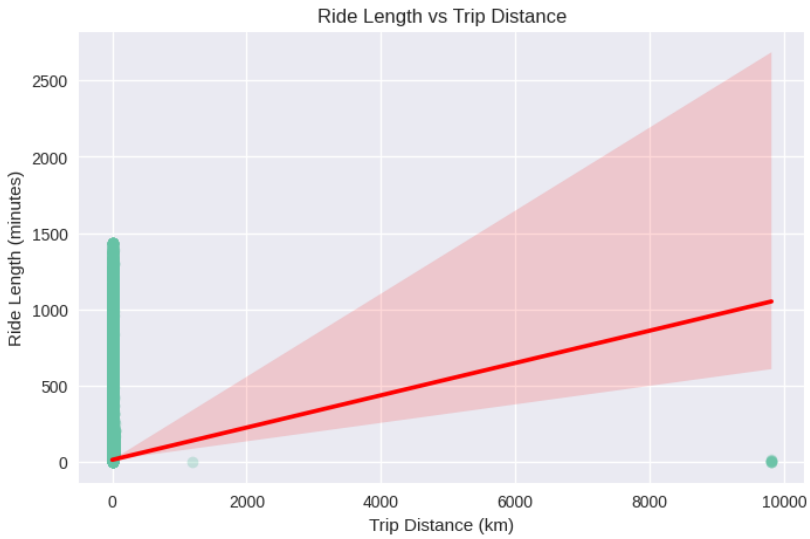
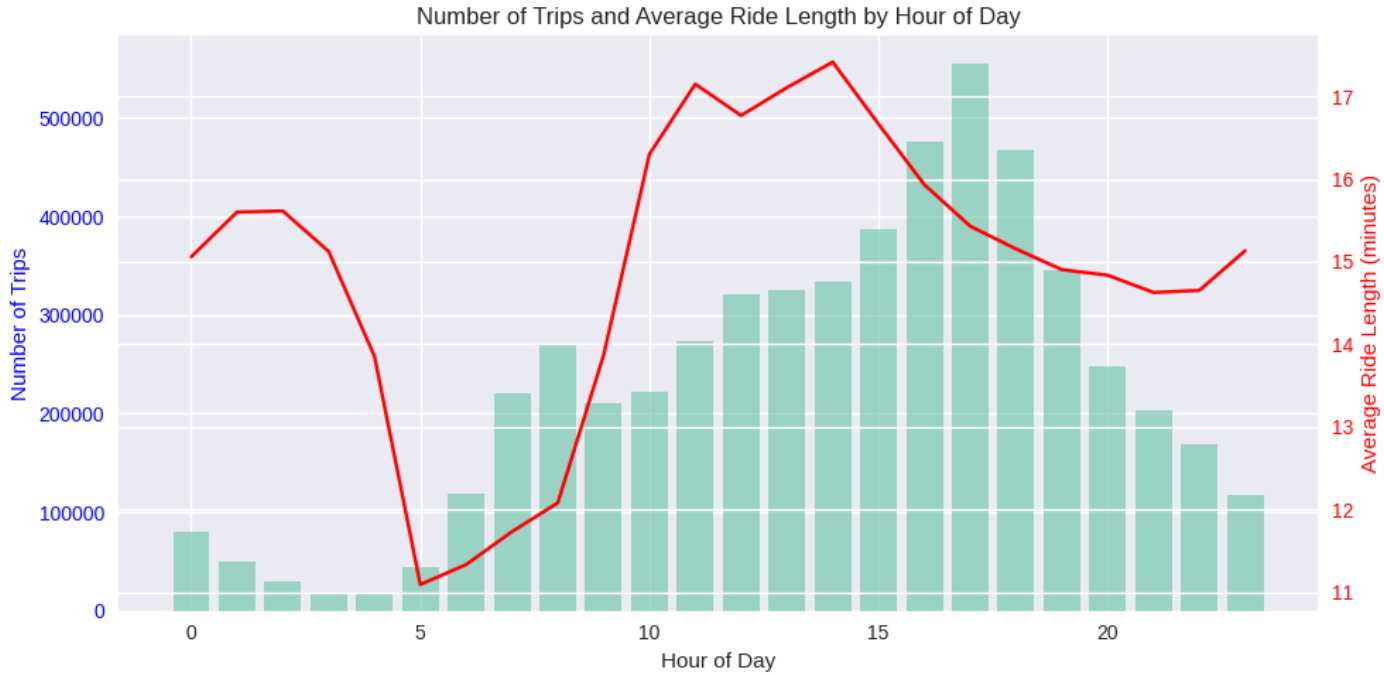
5. Correlation Analysis

1. **Ride Length vs Trip Distance**
   - Correlation = 0.047 → very low
   - Outlier filtering improved correlation slightly (0.038 → 0.047)
   - **Insight**: Most rides are short distances regardless of duration; idle trips or inconsistent behavior likely impact this metric.



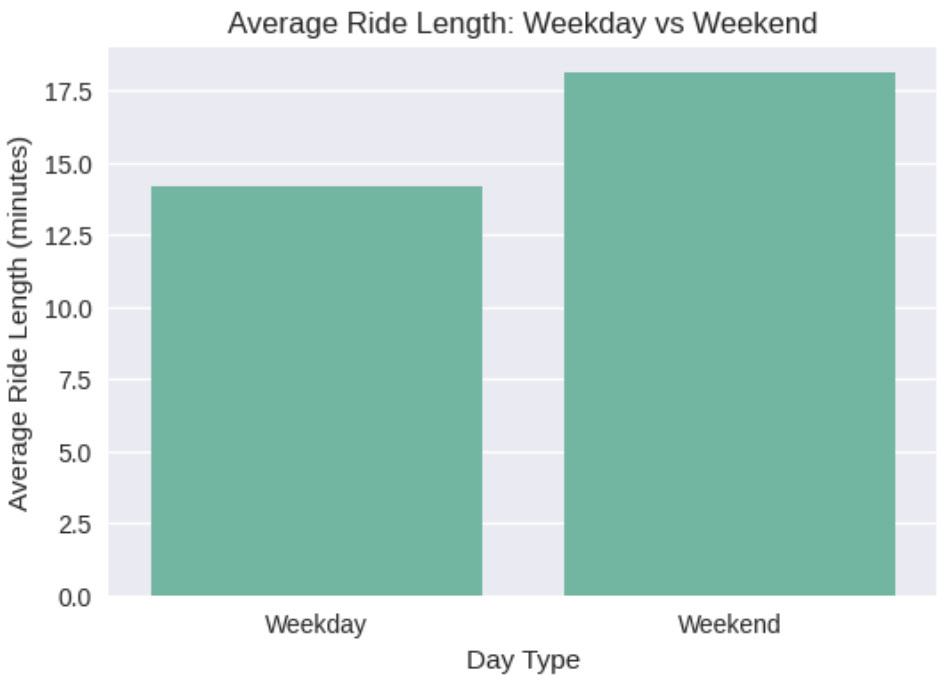
2. **Number of Trips vs Average Ride Length by Hour**
   - Correlation = 0.378 → weak positive relationship

   - **Insight**: Slightly longer rides tend to occur during busier hours, but operational planning should account for variability.



3. **Average Ride Length: Weekday vs Weekend**
   - Weekends: 18.1 min, Weekdays: 14.2 min

   - **Insight**: Longer weekend rides highlight recreational use; weekend resource allocation may require more flexible bike availability.



6. Key Insights

- **Temporal Usage Patterns:**
  - Peak rides occur on weekends and evenings; members dominate weekday commuting hours.

- **User Segmentation:**
  - Members are steady, short-duration riders; casual riders are seasonal, long-duration riders.

- **Station Demand:**
  - Streeter Dr & Grand Ave is consistently high-demand across all metrics; station-specific bike redistribution can optimize utilization.

- **Rideable Type Preference:**
  - Electric bikes are highly favored by casual riders; fleet planning can balance classic vs electric bike availability.

- **Operational Implications:**
  - Understanding hourly, daily, and monthly patterns supports resource allocation, bike redistribution, and targeted marketing campaigns.